Patient Education and Counseling 127 (2024) 108359

Contents lists available at ScienceDirect

Patient Education and Counseling

journal homepage: www.journals.elsevier.com/patient-education-and-counseling

# Prognostic understanding among advanced heart failure patients and their caregivers: A longitudinal dyadic study

Louisa Camille Poco [a], Ellie Bostwick Andres [a], Ishwarya Balasubramanian [a], Isha Chaudhry [a], Chetna Malhotra [a,b,*]

[a] Lien Centre for Palliative Care, Duke-NUS Medical School,169857, Singapore
[b] Health Services and Systems Research, Duke-NUS Medical School, 169857, Singapore

ARTICLE INFO

*Keywords:*
Actor-partner interdependence framework
Caregiver psychological distress
Caregiver and patient prognostic understanding
Heart failure

ABSTRACT

*Objectives:* To examine heart failure patients' and caregivers' prognostic understanding (PU) over time, and patient and caregiver factors associated with their own and each other's PU.

*Methods:* We used longitudinal dyadic data from Singapore, involving surveys with 95 heart failure patient-caregiver dyads every 4 months for up to 4 years. We assessed the association of PU with patient health status, caregiver psychological distress and caregiving hours using random effects multinomial logistic models, controlling for patient and caregiver characteristics.

*Results:* At baseline, half of patients and caregivers reported correct PU. Patient and caregiver variables were associated with their own and each other's PU. Patients with poorer functional well-being were less likely to report correct PU [Average Marginal Effects (95 % CI) 0.008 (0.002, 0.015)] versus incorrect PU. Greater caregiver psychological distress was associated with a lower likelihood of caregivers reporting a correct PU [−0.008 (−0.014, −0.002)]. Higher caregiving hours reduced the likelihood of patients reporting correct [− 0.002 (−0.003, −0.001)] and increased the likelihood of patients reporting uncertain [0.001 (0, 0.002)] PU.

*Conclusions:* We found PU among patients and caregivers was influenced by their own and each other's experience. Our findings highlight the importance of ongoing communication to enhance PU of patients and caregivers.

## 1. Introduction

Current clinical guidelines [1] advocate for providers to ensure that all patients with heart failure (HF) and their caregivers understand the disease trajectory and its prognosis. This understanding can facilitate shared decision-making between providers, patients and caregivers [2]. Existing research further underscores the significance of correct prognostic understanding (PU), showing that patients who lack this information are often treated more aggressively which may be in conflict with their goals and values [3–5]. Nevertheless, multiple studies report that many patients with HF have an incorrect PU [4,6–8], defined as incorrect understanding of HF's curability. Similarly, caregivers, despite playing an integral role in assisting patients with treatment decisions [2], frequently report an incorrect PU [9,10].

In addition to gaps in patient-physician communication and the unpredictable HF trajectory [3], patients' illness experiences also shape theirs and their caregivers' individual PU. Studies indicate patients with poorer health are more likely to accept their prognosis, thus reporting a correct PU.[11–14] Conversely, patients in better health may experience denial and report an incorrect PU [11]. Moreover, while cross-sectional studies have explored the association of caregivers' psychological distress and burden with patients' and caregivers' PU in context of advanced cancer, the directionality of association remains unclear due to conflicting and non-significant results [15–17]. In the context of HF patients and caregivers, no study has assessed this association. Notably, existing HF studies often overlook interdependence between patients and caregivers, a key aspect highlighted in a recent review on HF management [18].

The interdependence between patients and caregivers, stemming from their many shared experiences and outcomes, is a key component of the theory of dyadic illness management [19]. This theory advocates for a dyadic approach to better understand patient and caregiver outcomes. However, there is limited literature using this approach to investigate whether patients with HF and their caregivers influence each

* Correspondence to: Duke-NUS Medical School, 8 College Road, 169857, Singapore.
*E-mail address:* chetna.malhotra@duke-nus.edu.sg (C. Malhotra).

https://doi.org/10.1016/j.pec.2024.108359
Received 2 February 2024; Received in revised form 2 May 2024; Accepted 15 June 2024
Available online 17 June 2024

other's PU [18]. Furthermore, given advancements in HF treatments have prolonged patient life expectancy and improved quality of life, exploring PU longitudinally becomes essential.

Hence, our *first* aim was to use longitudinal data over 48 months, to assess PU among patients with HF and their caregivers, and the extent to which their perceptions change over time. Prior studies by us and others have shown that PU among patients with an advanced cancer changes over time [11,20]. We thus hypothesized that PU of both patients with HF and their caregivers would also change over time. Our *second* aim was to assess patient and caregiver factors associated with their own and with each other's PU using the Actor-Partner Interdependence (API) framework. This framework accounts for the interdependence within the dyads, estimating the association of one dyad member's characteristics with their own (actor effect) as well as with their partner's (partner effect) outcomes. As in previous studies [11–16], we hypothesized that when patients are in better health, patients and their caregivers would be less likely to report a correct PU. We also hypothesized that caregiver psychological distress and burden (measured by caregiving hours) would be associated with patients' and caregivers' PU, though direction of association remains unclear.

## 2. Data and methods

### 2.1. Sample

This paper uses data from the Singapore Cohort Study of Patients with Advanced Heart Failure (SCOPAH) [21]. We recruited 251 patients with advanced heart failure and their caregivers from two tertiary institutions in Singapore between July 2017 and April 2019. Eligible patients were Singaporean citizens or Permanent Residents, aged 21 years or older, with symptoms severe enough to be classified under New York Heart Association class 3 or 4. We excluded patients with major psychiatric disorders or who were cognitively impaired as assessed by the Montreal Cognitive Assessment test (score below 10). The study received approval from the SingHealth Centralised Institutional Review Board (2016/3046).

Eligible caregiver participants were aged 21 years and above and were the main or one of the main persons providing care to the patient *or* ensuring provision of care *or* involved in making patient treatment decisions. We excluded domestic helpers. We surveyed patients and caregivers every 4 months until the patients' death. Trained interviewers surveyed patients and caregivers in-person, on phone or through survey links. Survey data was captured on Qualtrics using electronic tablets. The sample size calculation for this cohort study is included in the published protocol [21]. Post-hoc power calculations showed the analytical sub-sample used had sufficient power for the conducted analyses.

The analytic sample for this study incorporates 4 years of follow-up data for dyads, encompassing up to 13 survey responses per dyad. For each survey wave, only dyads responding to the survey within a 2-week timeframe of each other were included.

### 2.2. Study measures

#### 2.2.1. Main outcome: prognostic understanding (PU)

At each time point, we asked patients and caregivers, "*Does your doctor think that all the treatments that you (patient) are currently receiving will help to cure your (patient's) heart condition?*" [6,7] Response options were yes, no, and not sure. We categorized PU into: (1) incorrect (yes), (2) uncertain (not sure), and (3) correct (no). In a sensitivity analysis, we dichotomized PU as either incorrect (yes or uncertain) or correct (no).

#### 2.2.2. Patient time-varying independent variables

Patient health status: We used the Functional Assessment of Cancer Therapy – General (FACT-G) subscale to assess patient physical and functional well-being. Each subscale comprises 7 items rated on a 5-point Likert scale (0−4) [22]. Total scores range from 0 to 28, with higher scores indicating greater physical or functional well-being. Past studies report concordance between FACT-G and Kansas City Cardiomyopathy Questionnaire, a validated quality-of-life measure among HF patients [23]. .

#### 2.2.3. Caregiver time-varying independent variables

Psychological distress: We measured caregiver psychological distress using the 14-item Hospital Anxiety and Depression Scale (HADS) [24]. Each item was rated on a 4-point scale (0−3). Total HADS scores range from 0–42, with higher scores indicating worse psychological distress.

Caregiving hours: Caregivers reported the average number of hours spent caring for or ensuring provision of care for the patient in a typical week, limited to a maximum of 16 h per day to account for sleep and rest time. A similar approach has been used in previous papers [25].

#### 2.2.4. Control variables

Following prior studies [15–17], we also controlled for time-varying patient and caregiver variables collected at each wave, and which could affect patient and caregiver PU, such as patient and caregiver age, whether caregiver provides unpaid care for others, patients' comorbidities (number of comorbidities indicated from a list of 10), if the patient is cared for by individuals other than the main caregiver, and caregivers' perceived relationship quality with the patients. We assessed caregivers' perceived relationship quality with the patients using a 4-item scale [26], with each item rated on a 4-point Likert scale (0−3). Responses were summed to calculate a total score (range: 0–12; higher scores indicating better relationship quality). The items elicited how emotionally close they felt to the patient, how was the communication between them and the patient, how similar their views were about life to those of the patient, and how well did they and the patient get along. We included time of assessment (survey wave) as a control variable to account for changes in the PU over time.

Table A1 presents the Cronbach alpha for the instruments included in the study (patient health status, caregiver psychological distress and quality of relationship), with the values for all measures indicating good internal consistency.

### 2.3. Statistical analysis

We first describe our analytical sample of 95 patient-caregiver dyads based on the patient and caregiver independent variables listed above. Then, we described the distribution of PU and changes in PU for 95 patients and caregivers separately at each time point from baseline to 48 months. To analyze changes in PU over time, we restricted the sample to patients and caregivers who reported PU in at least two surveys at consecutive time points (n = 72 dyads).

We used the full analytical sample of 95 patient-caregiver dyads that were surveyed two weeks apart to test the theory of dyadic illness management[19] and to implement the API framework [27]. (Figure A2) Random effects multinomial logistic regression models with dyad identification number specified as random intercept were estimated given the nominal categorical outcome variable (0 = incorrect PU, 1 = uncertain, 2 = correct). The model assumed uncorrelated unobserved individual effects with observed independent variables. Appendix A of the supplementary materials details the description of our dataset structure in order to implement the API framework.

We first estimated three separate regression models each controlling for a single independent variable of interest (patient health status, caregiver psychological distress and caregiving hours), without adjusting for covariates. Each regression included interaction terms between participant role (caregiver =1, patient=0) and the patient/caregiver time-varying independent variables of interest (patient health status, caregiver psychological distress, and caregiving hours) to estimate actor and partner effects. We refer to these results as unadjusted estimates.

We then incorporated all the independent variables of interest in a single regression model and adjusted for the control variables described above. We refer to these as adjusted estimates. For patients and caregivers, we investigated whether patients' health status, caregivers' psychological distress and caregiving hours were significant factors affecting patient or caregiver PU. We did so by estimating the average marginal effects (AME) which quantifies the average change in outcome for a one-unit change in the main independent variables.

We defined patient health status using two measures, functional and physical well-being, and these two variables are highly correlated with one another (Pearson's coefficient $\geq 0.5$). To prevent multicollinearity issue between these two independent variables, we included functional and physical well-being in separate models and estimated these regressions separately.

We also performed a sensitivity analysis to check the robustness of the main results. We use an alternative definition of PU, which is a binary variable to represent correct PU (=1) and uncertain or incorrect PU (=0). We estimated random effects logistic regression models incorporating the same independent and control variables as above.

We used Stata version 17 for all analyses. We used $P < 0.10$ to determine significance.

## 3. Results

The analytic sample included 95 eligible patient-caregiver dyads. (Figure A1) Most patients (72 %) were male, and most caregivers (80 %) were female. Patients and caregivers were, on average, aged 69 [SD:11] and 56 [SD:14] years, respectively. About two-thirds of caregivers were spouses of the patient. The average patient functional and patient well-being scores were 15.8 [SD: 5.6] and 20.1 [SD: 5.9] respectively. Meanwhile, 63 % of patients were cared for by individuals other than the caregiver. The average score for caregiver perceived quality of relationship with the patient was 9.5. Mean caregiver psychological distress was 7.6 [SD: 7.5]. Caregivers reported an average of 23 [SD: 28] caregiving hours in a typical week. (Table 1).

1. *Distribution of levels and changes in patient and caregiver PU over time*
2. *Patient and caregiver factors associated with PU*

Figs. 1a and 1b illustrate the distribution of patient and caregiver PU from baseline to 48 months. At baseline, half of patients and caregivers reported correct PU, however, over time, a higher proportion of patients than caregivers reported correct PU. Out of the 95 dyads, 72 (75 %) responded to at least two surveys at consecutive time points. Figs. 2a and 2b illustrate the change in patients' and caregivers' PU over time for this sub-sample of dyads. These figures show that patients' and caregivers' PU changed at each time point, towards both correct and incorrect PU. Out of the 72 dyads, on average, 21 % of both patients and caregivers changed their reported PU from the previous timepoint to uncertain/incorrect PU, and 14 % of patients and 15 % of caregivers changed their reported PU from the previous time point to correct PU. On average, 13 % and 11 % of patients' and caregivers' total responses showed a change in PU from the previous timepoint, with an average of 2 changes in PU per respondent over the study period.

Table 2 presents the estimation results for each of the three independent variables of interest estimated separately without adjusting for control variables (unadjusted estimates). The reference group for all the estimated multinomial logit models is incorrect PU. Poorer patient functional well-being was significantly associated with a reduced likelihood of patients reporting correct PU and an increased likelihood of patients and caregivers reporting uncertain PU (versus incorrect PU). Higher caregiver psychological distress was significantly associated with a decreased likelihood of caregivers reporting correct PU. Greater caregiving hours were significantly associated with a reduced likelihood of patients reporting correct PU and increased likelihood of patients reporting uncertain PU.

Table 3 presents adjusted estimates including all time-invariant and time-variant control variables and independent variables of interest in one regression model as described in the methods. The full estimation results of the multinomial logistic regression models are reported in Table A2 of the supplementary materials**.** Poorer patient functional well-being was also significantly associated with a lower likelihood of patients reporting correct PU [Average Marginal Effect (95 % Confidence Interval) 0.008 (0.002,0.015)] and a higher likelihood of both patients and caregivers reporting uncertain PU [patient: − 0.010 (−0.016, −0.005); caregiver: − 0.006 (−0.013,0)]. Higher caregiver psychological distress was significantly associated with a reduced likelihood of caregivers reporting correct (versus incorrect) PU [− 0.008 (−0.014,−0.002)]. More caregiving hours were significantly associated with lower likelihood of patients [− 0.002 (−0.003, −0.001)] as well as caregivers [− 0.001 (−0.003,0)] indicating correct PU, and a higher likelihood of patients reporting uncertain (versus incorrect) PU [0.001 (0,0.002)]. Table A3 confirms the robustness of main results using an alternative definition of patient health status, physical well-being, and controlling the same variables as in the previous specification. Poorer physical well-being was associated with a decreased likelihood of patients reporting correct PU and increased likelihood of reporting uncertain (versus incorrect) PU, while higher caregiver psychological distress was associated with a decreased likelihood of caregivers reporting a correct PU.

### 3.1. Sensitivity analyses

For the sensitivity analyses, we used correct PU as the outcome variable and controlled for the same variables as in the previous specification. Results as shown in Table A4 were also generally consistent with the main findings. For instance, poorer functional well-being (Model 1) was associated with a lower likelihood of patients reporting correct [0.009 (0.003, 0.015)] compared to incorrect/uncertain PU. Higher caregiver psychological distress was associated with a reduced likelihood of caregivers reporting correct (versus incorrect/uncertain) PU [− 0.007 (−0.013, −0.001)]. Higher caregiving hours were also associated with a lower likelihood of both patients [− 0.002 (−0.003, 0)] and caregivers [− 0.001 (−0.003, 0)] reporting correct compared to incorrect/uncertain PU.

**Table 1**
Patient and caregiver characteristics at baseline (n = 95 dyads).

| *Patient characteristics* | N (%) or Mean ± Std Deviation |
|---|---|
| Male | 68 (71.6) |
| Age | 68.61 ± 11.07 |
| Chinese ethnicity | 55 (57.9) |
| No or primary education | 50 (52.6) |
| Secondary/vocational education | 32 (33.7) |
| Junior college/Diploma/University education | 13 (13.7) |
| Number of comorbidities | 3.07 ± 1.74 |
| Functional well-being score [range: 0-28] | 15.76 ± 5.56 |
| Physical well-being score [range: 0-28] | 20.13 ± 5.94 |
| *Caregiver characteristics* | N (%) or Mean ± Std Deviation |
| Male | 19 (20.0) |
| Age | 56.12 ± 14.20 |
| Chinese ethnicity | 57 (60.0) |
| No or primary education | 23 (24.2) |
| Secondary/vocational education | 35 (36.8) |
| Junior college/Diploma/University education | 37 (38.9) |
| Relationship with patient - spouse | 55 (57.9) |
| If caregiver provides unpaid care to others | 48 (50.5) |
| Quality of relationship [range: 0-12] | 9.45 ± 2.73 |
| If patient receives additional help | 60 (63.2) |
| Psychological distress [range: 0-42] | 7.58 ± 7.51 |
| Caregiving hours in a typical week | 23.46 ± 28.14 |

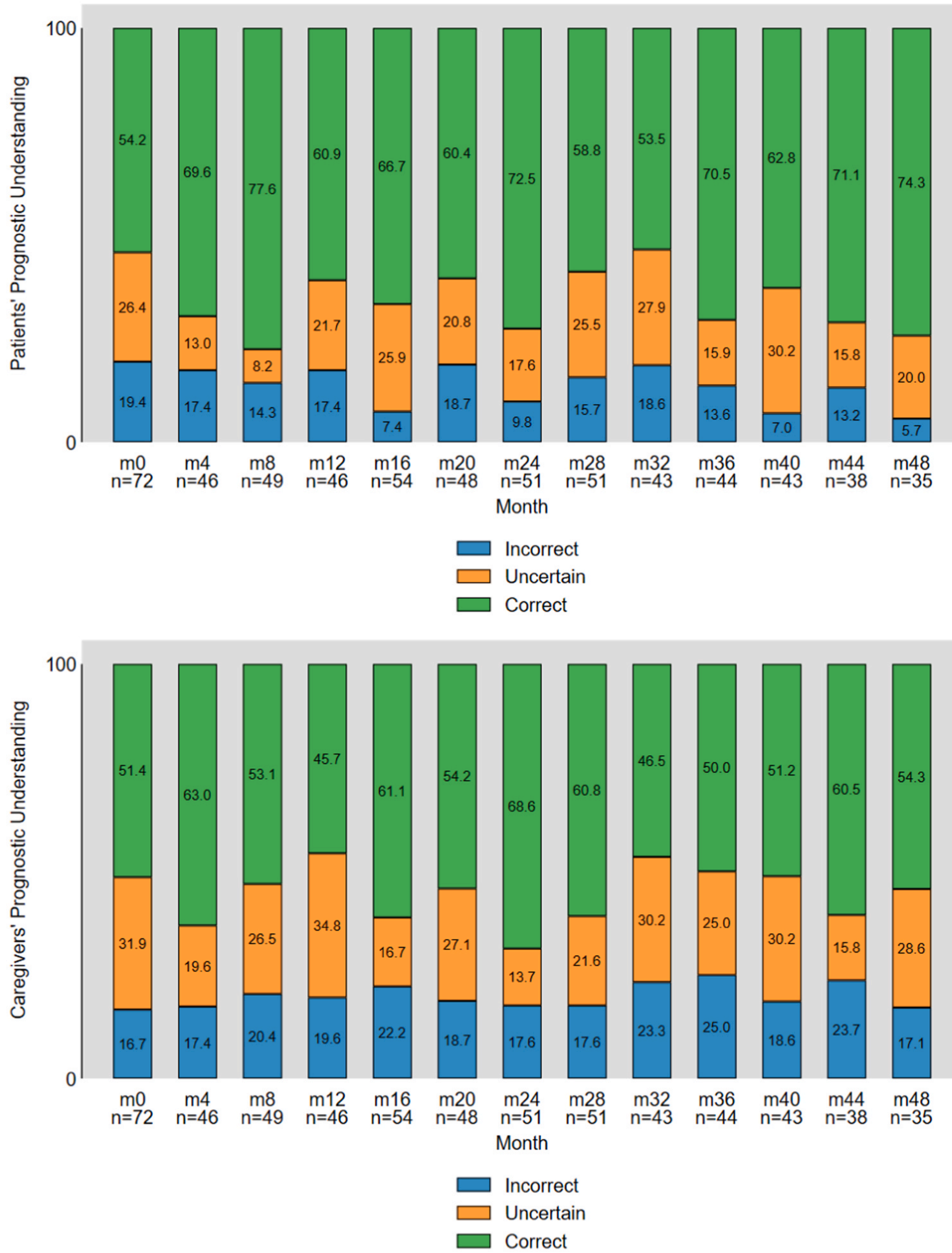


**Fig. 1.** a: Patients' prognostic understanding over time. Note: Based on a sample of n = 95 patients of patient-caregiver dyads. b: Caregivers' prognostic understanding over time. Note: Based on a sample of n = 95 patients of patient-caregiver dyads.

## 4. Discussion and conclusion

### 4.1. Discussion

We used 4 years of data encompassing up to 13 surveys from patients with HF and their caregivers to examine changes in PU. Despite the incurable nature of HF, patients are now living longer and thus, our study provides new understanding of how patients' and caregivers' PU evolves over time. We also used dyadic data to understand the extent to which patient and caregiver factors shape their own and each other's PU.

We found half of patients and caregivers in our sample reported correct PU at baseline. If we assume doctors communicated correct prognostic information to all patients and caregivers in line with clinical guidelines, we find discordance between doctor communication and patient/caregiver PU [8,28]. This could be due to optimism bias among patients and caregivers regarding their illness [29–31]. Alternatively, doctors may have hesitated to communicate prognosis clearly to patients and caregivers. Previous studies have noted the challenges doctors face in communicating prognosis clearly [3,9,32,33].

This study shows PU changed both towards and away from correct PU, with about 35 % of respondents changing their reported PU at each time point. These findings are notable in demonstrating PU is not a stable construct, and a sizeable proportion of respondents change from correct to uncertain/incorrect PU. The proportion of respondents with incorrect PU and whose PU changes is lower than those noted in previous studies of patients with advanced cancer and caregivers of persons with severe dementia [11,34]. One potential explanation for this difference is the way the PU question was framed. The current study asked respondents to report what they thought their doctors believed [7], while other studies have asked respondents to report their own prognostic beliefs. Framing the question around respondents' doctor's beliefs potentially reduced optimism bias frequently associated with reporting ones' own prognostic beliefs.


**Fig. 2.** a: Change in patients' prognostic understanding over time. Note: Based on a sample of n = 72 patients of patient-caregiver dyads. Each horizontal line in the figure represents a patient who reported PU at least two surveys at consecutive time points. b: Change in caregivers' prognostic understanding over time. Note: Based on a sample of n = 72 patients of patient-caregiver dyads. Each horizontal line in the figure represents a caregiver who reported PU at least two surveys at consecutive time points.

Our findings suggest that indeed patient health status influenced patient and caregiver PU. However, contrary to our hypothesis, a deterioration in patient health status was associated with patients being less likely to report a correct PU and more likely to report an uncertain PU, while caregivers were also more likely to report an uncertain PU. One possible explanation for this finding is that patients experience cognitive dissonance when the prognostic information they receive from their doctor does not align with their illness experience. As a result, patients use coping strategies such as denial and avoidance, leading them to confound their health status with their prognosis [11,30,35]. Likewise, caregivers too may experience cognitive dissonance, but to a lesser degree, making them more likely to report uncertain rather than incorrect PU. Similar findings have been previously reported among patients with advanced cancer and their caregivers [12,36].

**Table 2**
Factors associated with prognostic understanding of patients and caregivers (unadjusted estimates from multinomial logistic regression, n = 1240 observations)[a].

| Outcome (REF: Incorrect) | Independent Variables | Patient Prognostic Understanding | | Caregiver Prognostic Understanding | |
|---|---|---|---|---|---|
| | | Average Marginal Effect[2] (95 % CI) | p-value | Average Marginal Effect[b] (95 % CI) | p-value |
| Correct | Patient functional well-being | 0.008 (0.002, 0.014) | 0.009 | -0.003 (−0.009, 0.004) | 0.390 |
| | Caregiver psychological distress | -0.002 (−0.007, 0.004) | 0.547 | -0.006 (−0.011, 0.000) | 0.051 |
| | Caregiving hours | -0.002 (−0.003, 0.000) | 0.007 | -0.001 (−0.002, 0.000) | 0.132 |
| Uncertain | Patient functional well-being | -0.01 (−0.016, −0.005) | 0.000 | -0.006 (−0.011, 0.000) | 0.055 |
| | Caregiver psychological distress | 0.003 (−0.002, 0.008) | 0.249 | 0.004 (−0.001, 0.009) | 0.099 |
| | Caregiving hours | 0.001 (0.000, 0.002) | 0.029 | 0.000 (−0.001, 0.001) | 0.515 |

[a] Unadjusted regression model with only one independent variable.
[b] Average marginal effects (AMEs) from multinomial logistic random effects regression estimates.

We demonstrated caregiver psychological distress and higher caregiving hours were associated with a reduced likelihood of reporting correct PU. Extensive research has shown patients' and caregivers' negative emotions may hinder their understanding of complex medical information, such as prognosis, during medical consultations [37–39]. Thus, it is possible distressed and burdened caregivers may be truly unable to understand or process the prognostic information provided to them by physicians. Distressed or burdened caregivers are also more likely to use avoidance as a coping strategy [40,41], which in turn hinders their acceptance of the patient's prognosis, making them more likely report uncertain or incorrect PU.

As caregiving hours increased, we found patients were less likely to report correct and more likely to report uncertain PU. We speculate that when caregivers spend more time with patients, they may transmit their hope and optimistic beliefs to patients. For older and less educated patients [42], who also require more caregiving support, family caregivers, guided by strong Confucian beliefs in filial piety, may request treating doctors limit prognostic communication to patients with the goal of preserving patient hope and reducing psychological distress [43,44]. Previous studies in this context have documented this dynamic between caregivers and doctors [44,45].

The main strength of this paper is its use of 4 years of longitudinal data, encompassing up to 13 observations per dyad, to analyze the extent to which HF patients' and their caregivers' PU evolves over time and factors associated with PU. We also contribute to the growing dyadic research in HF by identifying patient and caregiver factors associated with their own and each other's PU. Our study also has limitations. First, there was attrition (range: 24–63 % of dyads) over time. Second, the generalizability of our findings beyond the Singapore context needs to be confirmed. Third, unlike many existing measures of PU that ask patients about their beliefs about curability, we asked patients to report what their doctor believes. This question was chosen to reduce potential cognitive bias influencing reported PU. However, patient and caregiver PU may have been influenced by other sources (e.g. family/friends, media, websites, etc.) not captured by our measure. Future research can confirm our findings using various measures of PU.

### 4.2. Conclusion

Using longitudinal dyadic data collected over a 4-year period, our study provides new understanding of prognostic beliefs held by patients with HF and their caregivers. We find PU among HF patients and caregivers changes over time and is influenced not only by their own but also by each other's experiences. These findings emphasize the importance of regular and open communication between physicians, patients, and caregivers to improve PU in HF.

### 4.3. Practice implications

Our study has practice implications. Given the gaps in patient and caregiver reported PU, there are opportunities to improve doctor-patient-caregiver communication about prognosis. Doctors should be trained to discuss prognostic information with patients and caregivers in a clear and empathic manner. Empathy can help reduce patient and caregiver psychological distress during consultations, thereby enabling improved processing and acceptance of the prognosis shared. Since patient health status was also associated with PU, these discussions should happen regularly to account for these changes. Specific interventions to support caregivers and reduce their distress may also potentially improve their acceptance of the patient's prognosis [35,46]. Future research should develop and evaluate HF-centric dyadic interventions to improve PU.

## Funding

This work was supported by the Health Services Research Grant, Ministry of Health, Singapore [NMRC/HSRG/0053/2016] and the Lien Centre for Palliative Care [LCPC-IN20-0001].

## CRediT authorship contribution statement

**Isha Chaudhry:** Writing – review & editing, Visualization,

**Table 3**
Factors associated with prognostic understanding of patients and caregivers (adjusted estimates from multinomial logistic regression model, n = 1234 observations)[a].

| Outcome (REF: Incorrect) | Independent Variables | Patient Prognostic Understanding | | Caregiver Prognostic Understanding | |
|---|---|---|---|---|---|
| | | Average Marginal Effect[2] (95 % CI) | p-value | Average Marginal Effect[b] (95 % CI) | p-value |
| Correct | Patient functional well-being | 0.008 (0.002,0.015) | 0.009 | -0.004 (−0.011,0.003) | 0.245 |
| | Caregiver psychological distress | 0.000 (−0.006,0.005) | 0.909 | -0.008 (−0.014,−0.002) | 0.013 |
| | Caregiving hours | -0.002 (−0.003,−0.001) | 0.005 | -0.001 (−0.003, 0.000) | 0.066 |
| Uncertain | Patient functional well-being | -0.01 (−0.016,−0.005) | 0.000 | -0.006 (−0.013,0) | 0.045 |
| | Caregiver psychological distress | 0.000 (−0.005,0.005) | 0.935 | 0.003 (−0.003,0.009) | 0.305 |
| | Caregiving hours | 0.001 (0.000,0.002) | 0.011 | 0.000 (−0.001,0.002) | 0.515 |

[a] Adjusted regression model controls for patient and caregiver gender, educational attainment, ethnicity, age, as well as caregiver relationship with patient, caregivers' perceived quality of relationship with patient, if caregiver provides unpaid care for other family members, patients total number of comorbidities, if patient receives additional help, and survey wave. Patient functional well-being is the patient health status variable in this regression model.
[b] Average marginal effects (AMEs) from multinomial logistic random effects regression estimates.

Methodology. **Ishwarya Balasubramanian:** Writing – review & editing, Visualization, Methodology. **Ellie Bostwick Andres:** Writing – review & editing, Visualization, Methodology. **Louisa Camille Poco:** Writing – review & editing, Writing – original draft, Visualization, Methodology, Formal analysis, Data curation, Conceptualization. **Chetna Malhotra:** Writing – review & editing, Writing – original draft, Visualization, Supervision, Methodology, Conceptualization.

## Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.pec.2024.108359.

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